
ConnectomeQuest: Certificate-Carrying Interaction under Costly Partial Observation

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Abstract

1 Agents acting under partial observation should not treat an inferred state as evi-
2 dence that an answer or action is supported. We introduce CONNECTOMEQUEST,
3 a claim–receipt–certificate interface that separates solution existence, charged in-
4 formation acquisition, and independent verification. Under matched access on
5 H01, MANC, and HemiBrain, no candidate ordering dominates: random branch
6 diversification is strongest on H01, whereas an Attribute second-hop ordering im-
7 proves one held-out HemiBrain comparison by 0.0213 (two-comparison Bonferroni
8 95% interval [0.0030, 0.0397]). Replacing only the target predicate while holding
9 the anchor, wiring, controller, checkpoint, and random seed fixed raises visible
10 target prevalence from 0.0060 to 0.1476 on MANC and from 0.0191 to 0.1535
11 on HemiBrain, demonstrating strong sensitivity of acquisition to semantic target
12 granularity. In separate audits, a policy-independent verifier accepts all 22,662
13 unmodified nonempty certificates and rejects all 13,800 constructed mutations. In
14 RGB UnlockPickupDist, a prospectively frozen CPU confirmation on 221 eligible
15 long-plan layouts finds identical authorization decisions while dependency-indexed
16 validation reduces validation-and-replanning CPU time by 96.7% (layout-bootstrap
17 95% interval [96.4%, 96.9%]) relative to complete validation under four or eight
18 irrelevant receipt withdrawals; this endpoint includes index costs but excludes
19 perception, simulation, and action execution. After final support loss, checking
20 methods make zero unsupported authorizations, whereas unchecked reuse makes
21 511 of 513. Together, these results establish an access-controlled neuro-symbolic
22 protocol for acquiring, validating, and selectively revising evidence across graph
23 and RGB agents.

24 1 Introduction

25 An agent using a partially observed semantic map must distinguish what it has observed, what it
26 infers, and which acquired facts support its next decision. We organize budgeted semantic search
27 around three questions: whether a solution exists, what acquiring its evidence costs, and whether
28 another component can verify the resulting certificate. We call an interaction certificate-carrying
29 when every accepted answer is accompanied by a certificate assembled from environment-issued
30 receipts and accepted by a separately trusted verifier under the current query and episode context.

31 An observation can change confidence without improving affordable continuation actions: informa-
32 tion, decision value and witness validity are distinct quantities.

33 The principal contribution of CONNECTOMEQUEST is a certificate-carrying interaction protocol
34 and its evaluation procedure; learned rankers and symbolic planners are replaceable instantiations,
35 not the claimed protocol contribution. For any proposed output u , the task defines a finite set $\mathcal{E}(u)$
36 of required claims. Acceptance requires an active receipt supporting every required claim under a

Table 1: Notation used in the formal contract.

Symbol	Meaning
$G_d = (\mathcal{V}_d, \mathcal{R}_d, E_d, w_d)$	dataset graph and nonnegative edge weights
V, B, H	visibility-page size, action budget, and horizon
$q = (a, y); \omega = (m, x)$	query and graph witness
$\tau; P; \chi$	receipt; certificate; trusted verification context
$\mathcal{E}(u)$	finite set of required claims for output u
$s_t; z_t$	environment state and controller state
$\mathcal{K}_\pi; \pi$	controller instantiated with candidate-ordering policy π
$D_i \subseteq \mathcal{I}; K_i \subseteq \mathcal{K}$	receipt and semantic-key dependencies of plan step i
$\Delta_t = (W_t, N_t); \mathcal{J}(\Delta_t)$	typed evidence update; affected plan positions
$\Gamma_d(s, q)$	admissible API actions at environment state s
$\tilde{S}(\mathcal{K}_\pi; d, \mathcal{Q}_d, B, H)$	finite-sample certificate-success rate

37 trusted context. The graph verifier accepts final answers; the RGB verifier authorizes actions under
 38 the same typed interface. The graph benchmark then measures how semantic targets and acquisition
 39 rules affect search independently of checking; the RGB study instantiates the same required-claim
 40 interface with separate policies and predicates.

41 Connectomes provide a controlled setting for this problem: weighted synaptic wiring is combined
 42 with cell-type, region, and layer annotations. HemiBrain covers the adult fly central brain [15],
 43 MANC the male fly ventral nerve cord [18], and H01 a human cortical sample [16]. Their different
 44 degree, weight, and annotation distributions test whether an action-selection policy transfers across
 45 reconstructions and semantic schemas.

46 Full-graph query-answering and relational-search methods learn geometric or path representations
 47 [5, 8, 14]; our protocol instead charges observation and requires acquired evidence. Costed evaluation
 48 and value-of-observation planning study which observation to buy [6, 9, 12]. In embodied settings,
 49 active semantic mapping and task–motion planning couple perception, state estimation, and action
 50 constraints [1, 10]. Truth maintenance, incremental execution, and plan repair use dependency
 51 structure to localize revision [4, 7, 17, 19]. Provenance and proof-carrying code distinguish derivation
 52 records from consumer checks [11, 13]. We combine these established ideas in a controlled acquisition
 53 benchmark rather than claim a new general provenance formalism or universal planner.

54 The complete graph remains private to the environment. An agent pays for observation and traversal,
 55 then submits acquired evidence before its budget expires. This isolates action-conditioned observation,
 56 precondition enforcement and limited acquisition, also relevant to embodied semantic maps. The
 57 graph abstraction excludes low-level locomotion and manipulation; exact ground truth supports
 58 controlled acquisition and certificate audits.

59 Our contributions are:

- 60 • an explicit certificate-carrying interaction interface and a three-connectome benchmark for budgeted
 61 acquisition of semantic witnesses;
- 62 • a controller-matched held-out evaluation of six observable orderings, plus a paired target-granularity
 63 intervention that holds wiring fixed;
- 64 • a policy-independent verifier and mutation audit that isolate certificate integrity from policy success;
- 65 • a prospectively frozen RGB study showing that dependency-indexed validation preserves complete-
 66 validation decisions while reducing specified validation-and-replanning CPU time on long plans
 67 with sparse updates and preventing unsupported cached actions.

68 2 Active semantic connectome reasoning

69 **Common contract.** For task instance r , let $\mathcal{T}_r$ be its receipt space and $\mathcal{X}_r$ its trusted-context space.
 70 Define $\mathcal{C}_r = (\mathcal{U}_r, \mathcal{F}_r, \mathcal{E}_r, \text{Issue}_r, \text{Verify}_r)$, where $\mathcal{U}_r$ is the output space, $\mathcal{F}_r$ is the set of atomic
 71 claims, and $\mathcal{E}_r : \mathcal{U}_r \rightarrow \mathcal{P}_{\text{fin}}(\mathcal{F}_r)$ maps each output to its finite set of required claims. Issuance is the
 72 trusted relation $\text{Issue}_r \subseteq \mathcal{X}_r \times \mathcal{F}_r \times \mathcal{T}_r$; $(\chi, f, \tau) \in \text{Issue}_r$ means that the environment issued receipt
 73 τ for claim f in context χ . Each context also specifies the currently active receipt set $\text{Act}_r(\chi) \subseteq \mathcal{T}_r$.
 74 We say that τ supports f under χ exactly when $\tau \in \text{Act}_r(\chi)$ and $(\chi, f, \tau) \in \text{Issue}_r$. The verifier has
 75 type $\text{Verify}_r : \mathcal{U}_r \times \mathcal{T}_r^* \times \mathcal{X}_r \rightarrow \{0, 1\}$, where $\mathcal{T}_r^*$ is the set of finite receipt sequences. It accepts

76 exactly when every required claim has support in the submitted certificate:

$$\text{Verify}_r(u, P; \chi) = 1 \iff \forall f \in \mathcal{E}_r(u) \exists \tau \in P : \tau \text{ supports } f \text{ under } \chi.$$

77 The graph and RGB tasks below instantiate this interface; their policies, perception models, planners,
78 and required-claim definitions are task-specific.

79 **Graph, visibility, and query.** For dataset d , let $G_d = (\mathcal{V}_d, \mathcal{R}_d, E_d, w_d)$ be a directed weighted
80 multirelational graph, with $E_d \subseteq \mathcal{V}_d \times \mathcal{R}_d \times \mathcal{V}_d$ and $w_d : E_d \rightarrow \mathbb{R}_{\geq 0}$. Wiring uses relation
81 $p = \text{presynaptic_to}$; semantic evidence uses the dataset-specific relation ρ_d : has_type for the
82 fly graphs and in_cortical_layer for H01. For node u , relation r , page size V , and offset h , the
83 visibility operator $\text{Vis}_{d,r,V,h}(u)$ returns entries $h+1, \dots, h+V$ from the outgoing triples, sorted
84 by decreasing weight and then numeric destination ID. Write $\text{Vis}_{d,r,V}(u) = \text{Vis}_{d,r,V,0}(u)$. Each
85 subsequent $\text{INSPECT}(u, r)$ advances h for (u, r) ; changing focus does not.

86 For anchor neuron a and semantic target y , define $q = (a, y)$ and

$$A_d(a, y) = \{x \in \mathcal{V}_d : \exists m \in \mathcal{V}_d, (a, p, m), (m, p, x), (x, \rho_d, y) \in E_d\}. \quad (1)$$

87 A graph witness is $\omega = (m, x)$, with answer projection $\text{ans}(\omega) = x$, and its required claims are

$$\mathcal{E}_d(q, \omega) = \{(a, p, m), (m, p, x), (x, \rho_d, y)\}. \quad (2)$$

88 The task is to return x with acquired evidence for every member of $\mathcal{E}_d(q, \omega)$. Degenerate paths
89 are allowed only when every displayed directed edge occurs. Queries are generated at $V_{\text{gen}} = 32$
90 with one witness in the relevant first pages; $V < 32$ is therefore a visibility stress test without that
91 guarantee.

92 **Receipts and verification.** A graph receipt is $\tau = (\kappa, e, h_d, \text{id}(q), \eta, t, \iota)$, containing polarity
93 $\kappa \in \{+, -\}$, triple e , graph digest h_d , query identifier, episode nonce η , issuance step t , and receipt
94 identifier ι . The trusted context is $\chi = (h_d, q, \eta, t_{\text{cur}}, \mathcal{R})$, where registry $\mathcal{R}$ is a partial map from
95 issued identifiers to canonical payload digests. For a fixed canonical serialization and digest function
96 H , define $\text{ValidReceipt}(\tau; \chi) = 1$ exactly when its graph, query, and episode bindings match χ ,
97 $0 \leq t \leq t_{\text{cur}}$, and its digest equals $\mathcal{R}(\iota) = H(\text{payload}(\tau))$; an unregistered identifier is invalid. For
98 proposed answer x , the submission contains a graph witness $\omega = (m, x)$. A graph certificate P is
99 a finite receipt list, and $\text{Verify}_{d,q}(\omega, P; \chi) = 1$ exactly when P contains a valid positive receipt for
100 every member of $\mathcal{E}_d(q, \omega)$. Verification does not read G_d but is conditional on issuer-attested receipt
101 truth and the trusted registry.

102 **Proposition 1** (Conditional graph-certificate soundness). *If G_d is immutable and the issuer marks
103 a receipt positive only for an edge in E_d , then $\text{Verify}_{d,q}(\omega, P; \chi) = 1 \Rightarrow \text{ans}(\omega) \in A_d(a, y)$.
104 The accepted receipts instantiate Eq. (1). This does not imply certificate completeness, third-party
105 authentication, or correctness of a learned RGB interpretation.*

106 **Controlled process and controller.** The environment is the deterministic partially observed con-
107 trolled process

$$\mathcal{M}_{d,q} = (\mathcal{S}_{d,q}, \mathcal{A}_d, \mathcal{O}_d, T_{d,q}, \Omega_{d,q}, c_{d,q}, B, H, \mu_{0,d,q}),$$

108 where $T_{d,q} : \mathcal{S}_{d,q} \times \mathcal{A}_d \rightarrow \mathcal{S}_{d,q}$ is total, $\Omega_{d,q} : \mathcal{S}_{d,q} \rightarrow \mathcal{O}_d$ is the observation map, and $c_{d,q} :$
109 $\mathcal{S}_{d,q} \times \mathcal{A}_d \rightarrow \mathbb{R}_{\geq 0}$ is the charged cost, B the cost budget, H the horizon, and $\mu_{0,d,q}$ the initial-state
110 law. Admissible actions form $\Gamma_d(s, q) \subseteq \mathcal{A}_d$; an inadmissible attempt yields an API-rejection
111 transition and still incurs its stated cost. Environment state s_t contains the hidden graph, focus, issuer
112 state, pagination offsets, step, and remaining budget. Controller state z_t contains policy memory and
113 derived frontier state. For a candidate-ordering policy π , controller $\mathcal{K}_\pi$ and a fixed random tape ξ , let
114 $F_{\mathcal{K}_\pi}$ denote the controller-state update. They induce

$$\begin{aligned} o_t &= \Omega_{d,q}(s_t), & a_t &= \mathcal{K}_\pi(z_t, o_t; \xi), \\ s_{t+1} &= T_{d,q}(s_t, a_t), & z_{t+1} &= F_{\mathcal{K}_\pi}(z_t, o_t, a_t, \Omega_{d,q}(s_{t+1}); \xi). \end{aligned}$$

115 Thus π orders controller-supplied candidates; it cannot alter admissibility, observation, cost, or
116 certificate acceptance.

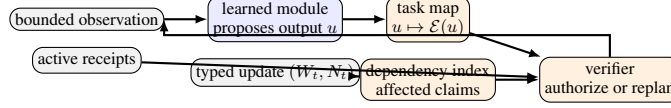


Figure 1: Common runtime loop. In the graph task, u is an answer and $\mathcal{E}(u)$ contains three required edge claims. In RGB planning, u is an action and $\mathcal{E}(u)$ contains its required symbolic claims. Policies, predicates, and learned weights are task-specific; the shared object is the certificate-carrying interaction protocol.

Actions and evaluation. The evaluated controller uses INSPECT, TRAVERSE, CHECKEDGE, BACKTRACK, and SUBMIT. CHECKEDGE is admissible only for a semantic triple (x, ρ_d, y) when x is the focus and an acquired two-hop wiring path ends at x . Every attempted non-submit action costs one; SUBMIT costs zero and consumes one step. The primary controller fixes $V = 32$, $B = H = 64$, and at most four answer checks per opened middle node.

For fixed query set $\mathcal{Q}_d = \{q_1, \dots, q_n\}$, let $\mathcal{L}_\pi$ be the fixed evaluation-seed set for policy π , with a singleton set for a deterministic policy. Let $Y_{i\ell} = 1$ iff controller $\mathcal{K}_\pi$ submits within B, H an answer whose graph certificate is accepted for q_i under seed $\ell \in \mathcal{L}_\pi$, and 0 otherwise. The reported statistic is

$$\hat{S}(\mathcal{K}_\pi; d, \mathcal{Q}_d, B, H) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|\mathcal{L}_\pi|} \sum_{\ell \in \mathcal{L}_\pi} Y_{i\ell}.$$

It is a finite-sample certificate-success rate conditional on the named controller, queries, checkpoints, and seeds, not an unspecified population probability.

3 Benchmark implementation

3.1 Learned candidate ordering

The Attribute scorer receives no numeric entity IDs or entity embeddings. Its two MLP heads rank each supplied hop-specific candidate list from 31 observable coordinates: acquired counts and weights, memory/frontier flags, within-list ranks, budget state, semantic matches, and paid-probe statistics when available. It is trained only on source graphs for 20 epochs with AdamW at 3×10^{-4} . Numeric-ID tie-breaking can still change visibility and rank-derived features; complete features, loss, sample counts, and the relabeling counterexample are in the supplement.

3.2 Controller and trust boundary

The fixed controller instantiates Eq. (1) as a query automaton. It inspects the anchor, orders the visible middle nodes, traverses to a selected middle node, inspects its successors, checks legal semantic claims, and backtracks on failure. Learned scores only permute candidate lists; the controller owns focus, action masking, budget, and the required-claim set. The general graph API permits more actions than this automaton, so the reported protocol is defined by controller $\mathcal{K}_\pi$ as well as API legality.

4 Experimental design

Data. We construct compact neuron-level KGs from official releases and exclude raw EM imagery, skeletons, meshes, and per-synapse point tables; the supplement reports the exact node and edge counts by relation type. H01 annotations are official cortical-layer labels sampled at soma centroids. MANC retains neurons with a non-null curated mancType; HemiBrain retains traced, uncropped neurons exported through neuPrint [3]. Inverse wiring edges are materialized. We use publicly released datasets under their respective research licenses; the supplement records source versions and access metadata.

First-page target prevalence. Define the *first-page two-hop candidate union*

$$X_{d,V}^{\text{fp}}(a) = \{x \in \mathcal{V}_d : \exists m, (a, p, m) \in \text{Vis}_{d,p,V}(a), (m, p, x) \in \text{Vis}_{d,p,V}(m)\}.$$

This set is computed offline by opening the first continuation page for every middle node disclosed on the anchor’s first page. It is a dataset diagnostic, not an agent observation, and no evaluated policy receives it. For query (a, y) , its visible target prevalence is

$$s_{d,V}(a, y) = \frac{|A_d(a, y) \cap X_{d,V}^{\text{fp}}(a)|}{\max\{1, |X_{d,V}^{\text{fp}}(a)|\}}.$$

A smaller value means that the target selects a smaller fraction of this offline candidate union. Target counts and answer-set sizes are accompanying statistics, not alternative definitions. Test queries use all seven H01 layer targets, 1,945 MANC type targets among 3,893 graph types, and 2,278 HemiBrain type targets among 5,554 graph types. Mean answer-set sizes are 45.5, 7.3, and 20.7, respectively. Cross-graph differences also include topology, weights, and annotation coverage; we do not attribute them to target prevalence alone.

Splits and evaluation blocks. Neuron anchors are assigned to train, validation, or test by seeded cryptographic hash; semantic targets are shared ontology nodes, not held-out neurons. For each leave-one-connectome-out direction, the Attribute and Recurrent policies train on the other two graphs with seeds 17, 29, and 43 and receive no entity embeddings. Hyperparameters and selector checkpoints use source validation only.

The main comparison is the prospectively frozen held-out-pair evaluation in Table 2. It holds the query automaton, visibility operator, budget, action restrictions, verifier, and query set fixed while changing only candidate ordering. Its controller ranks the complete visible first-hop frontier without a paid probe. The 5,000-pair target-coarsening block holds the anchor, wiring, visibility rule, controller, policy checkpoint, and random seed fixed while replacing only the target predicate; because that predicate changes the query, subsequent actions may differ. This is an explicitly exploratory mechanism audit.

Baselines and statistical unit. All main-table policies use observable information only. Random applies a deterministic permutation keyed by run seed, query ID, current node, and candidate ID. The Recurrent baseline uses the same 31 observed features and source-graph decision snapshots as the Attribute scorer, but trains an LSTM ranker with REINFORCE; its state resets at each ranking call. It is therefore an access-matched recurrent control, not a reproduction of full-graph MINERVA. The supplement gives architectures, training volumes, and hyperparameters.

Certificate success is averaged over 1,000 fixed query IDs and, for stochastic policies, three fixed training/action seeds. After an audit found repeated first anchors, we replaced query-wise resampling by a paired cluster bootstrap; this post-outcome correction changes uncertainty estimates, not policies or point estimates. It first averages seed outcomes within query, then resamples first-anchor clusters 10,000 times and retains every query in each sampled cluster. H01 has 837 clusters, MANC 807, and HemiBrain 806 (maximum cluster size four). The two planned fly-graph contrasts between Expert selector and Weight use Bonferroni-adjusted 97.5% intervals. H01 and all other contrasts are descriptive. The correction accounts for repeated anchors but remains conditional on fitted checkpoints, fixed seeds, the controller, and these graphs.

5 Results

5.1 Held-out evaluation under fixed access

We fixed the controller, page size $V = 32$, budget and horizon $B = H = 64$, candidate cap, policy checkpoints, and two fly-graph comparisons before evaluating 1,000 previously unused query pairs per graph. The queries share no anchor–target pair with earlier evaluation sets. Because first anchors repeat, confidence intervals resample first-anchor clusters. The three action/model seeds are fixed computational replications; the intervals are conditional on these seeds and on this query population.

Weight orders candidates by observed synaptic weight. Random uses a query- and seed-keyed random permutation. Recurrent is an access-matched LSTM policy trained with REINFORCE. Attribute is the two-head listwise MLP in Section 3. Expert selector chooses one of the observable policies from source-graph features before acting. Random→Attribute randomizes only the first hop and uses Attribute ordering at the second. Every policy receives the same observations, action costs, and certificate-verification rule.

Table 2: Prospectively frozen evaluation on 1,000 previously unused query pairs per graph. Entries are certificate success/unconditional charged actions, averaged across seeds 17/29/43 for stochastic policies; Weight is deterministic. All policies use the same frozen controller and observations.

Graph	Weight	Random	Recurrent	Attribute	Expert selector	Random→Attribute
H01	0.4090/42.63	0.5100/39.82	0.4840/40.64	0.4117/41.27	0.3920/43.14	0.4017/42.05
MANC	0.0990/60.18	0.1050/60.11	0.1030/60.01	0.1107/59.96	0.1120/60.03	0.1003/60.19
HemiBrain	0.1270/59.52	0.1347/59.23	0.1437/58.86	0.1343/58.91	0.1483/58.83	0.1367/58.86

No policy dominates on all three graphs. Relative to Weight, Expert selector changes certificate success by +0.0213 on HemiBrain (two-comparison Bonferroni 95% interval $[0.0030, 0.0397]$), +0.0130 on MANC ($[-0.0094, 0.0360]$), and -0.0170 on H01 (descriptive nominal interval $[-0.0295, -0.0048]$). Random is strongest on H01, whereas Expert selector is strongest on HemiBrain. A matched hop-attribution audit shows that replacing the selector’s first hop by Weight leaves all 9,000 outcomes unchanged. Thus the observed HemiBrain difference is attributable to second-hop ordering, not to the first-hop selector. Accordingly, the selector is an evaluated component rather than the primary contribution.

Raw action means can reward early failure. Outcome-conditioned aggregation shows the opposite: unsuccessful episodes average about 51 charged actions on H01 and 63 on MANC and HemiBrain. The controller reaches its 63-action exploration limit in 29–37% of H01 episodes, 89–90% of MANC episodes, and 85–87% of HemiBrain episodes. The supplement gives every policy; these data identify acquisition exhaustion rather than low-cost failure.

5.2 Target granularity controls acquisition difficulty

Let Y_d be the native target set and $\bar{Y}_d = \{\bar{y}_1, \dots, \bar{y}_7\}$ a disjoint set of synthetic coarse targets. A fixed map $g_d : Y_d \rightarrow \bar{Y}_d$ replaces each semantic edge (x, ρ_d, y) by $(x, \bar{\rho}_d, g_d(y))$. Consequently $A_d(a, y) \subseteq \bar{A}_d(a, g_d(y))$. This set inclusion is elementary; the experiment measures its effect on paired anchors under a fixed visibility rule, controller, policy checkpoint, and random tape. The target predicate, and hence the realized actions, may differ.

We apply the transformation to 5,000 paired MANC and HemiBrain instances while holding wiring, anchor, pairing identifier, policy checkpoint, and random seed fixed. Visible target prevalence

Table 3: Paired label-coarsening audit, 5,000 original native queries per fly graph. Every entry is native/coarse. Visible target prevalence is the relevant fraction among first-page two-hop candidate nodes; other columns are Success(64), averaged over the original action/model seeds. No new model fitting.

Graph	Visible target prevalence	Weight	Random	Attribute
MANC	0.0060/0.1476	0.0972/0.8226	0.1093/0.9036	0.1088/0.8644
HemiBrain	0.0191/0.1535	0.1290/0.7858	0.1765/0.8887	0.1373/0.8167

increases from 0.0060 to 0.1476 on MANC and from 0.0191 to 0.1535 on HemiBrain; success rises substantially for all three reported policies. Thus semantic target granularity materially changes acquisition difficulty in this controlled intervention; it is not evidence that one ordering algorithm alone solves the native regime. A privileged visible-witness ordering solves every generated query within 16 actions, confirming that the gap is observable acquisition rather than answer non-existence.

5.3 Certificate verification

The policy-independent standard-library verifier sees the public query, proposed witness, certificate, and trusted issuance context, but no graph or answer set. It accepts all 22,662 unmodified nonempty submissions from 60,000 graph-search episodes; empty submissions are not certificates. In a separate diagnostic it rejects all 13,800 constructed binding and structural mutations. This tests policy-independent logical support under a trusted issuer, not search quality or an attack-rate bound; class-by-graph counts and the trusted-registry assumption are in the supplement.

6 RGB semantic-map evaluation

BabyAI UnlockPickupDist [2] instantiates the same claim–receipt–certificate interface. Every method receives the same egocentric 7×7 RGB image, position relative to the initial pose, exact cardinal direction, inventory type, mission string, and last-action outcome. A frozen tile classifier converts the RGB image into semantic-map claims, each linked to the corresponding immutable image receipt. For a proposed action a , $\mathcal{E}_t(a)$ is its finite set of required symbolic claims. Authorization requires active receipt support for every claim in $\mathcal{E}_t(a)$. A receipt proves image acquisition, not the physical truth of the learned interpretation.

Dependency-indexed plan validation. Let $\mathcal{I}$ be the receipt-identifier set and $\mathcal{K}$ the set of semantic update keys. At decision time t , the deterministic planner’s retained suffix is $\Pi_t = (a_{c_t}, \dots, a_L)$. Let $A_t \subseteq \mathcal{I}$ be the active receipts and let θ_t be the current semantic-key valuation. For every $i \geq c_t$, define $\text{Supp}_i(A_t, \theta_t) \in \{0, 1\}$ to indicate whether every required claim of a_i has support from an active receipt.

The recorded set $D_i \subseteq \mathcal{I}$ is *receipt-complete* when the following implication holds for all A, A' and θ : if A and A' differ only in receipt r and $\text{Supp}_i(A, \theta) \neq \text{Supp}_i(A', \theta)$, then $r \in D_i$. The recorded set $K_i \subseteq \mathcal{K}$ is *key-complete* when it contains every semantic key whose value is read either by Supp_i or by the planner when constructing a_i and its required-claim set. A changed key in K_i therefore marks the retained suffix stale even if its currently cached support value remains one. For an update from (A_t, θ_t) to (A_{t+1}, θ_{t+1}) , let $W_t = A_t \triangle A_{t+1}$ and let N_t contain exactly the keys whose values changed; write $\Delta_t = (W_t, N_t)$. The affected positions are

$$\mathcal{J}(\Delta_t) = \{i \geq c_t : D_i \cap W_t \neq \emptyset \text{ or } K_i \cap N_t \neq \emptyset\}.$$

The dependency index stores the exact inverse maps $L_{\mathcal{I}}(r) = \{i : r \in D_i\}$ and $L_{\mathcal{K}}(k) = \{i : k \in K_i\}$.

Proposition 2 (Coverage and authorization). *If D_i is receipt-complete, K_i is key-complete, and both inverse maps are exact, then $\mathcal{J}(\Delta_t)$ contains every suffix position whose support value changes and every position whose recorded planner-read key changes. Consequently, a controller that revalidates those positions, replans after a recorded planner-read change, and authorizes the next action only with current support never authorizes that checked action after its final active support is withdrawn. The conclusion is relative to the trusted issuer and symbolic action model; it does not establish physical correctness.*

Proposition 3 (Selective-validation equivalence). *Suppose complete suffix validation and dependency-indexed validation receive the same retained suffix, active receipts, semantic valuation, update, and deterministic planner, and an exact cached pre-update support vector. Both procedures use the same retention rule: retain exactly when all post-update support values equal one and $N_t \cap \bigcup_{i=c_t}^L K_i = \emptyset$; otherwise invoke the common planner. Under the completeness assumptions of Proposition 2, they therefore return the same next authorization decision. This statement does not compare either procedure with eager replanning after a change outside the recorded read dependencies that could make an unrelated alternative plan preferable.*

Corollary 1 (Amortized location work). *Within one retained-plan epoch with fixed cursor c , let $Z = \sum_{i=c}^L (|D_i| + |K_i|)$ at index construction. With expected-constant-time hash lookup, construction followed by M updates has location work*

$$O\left(Z + \sum_{t=1}^M \left[|W_t| + |N_t| + \sum_{r \in W_t} |L_{\mathcal{I}}(r)| + \sum_{k \in N_t} |L_{\mathcal{K}}(k)| \right]\right).$$

Complete suffix validation instead visits $M(L - c + 1)$ positions. Across multiple retained-plan epochs, both costs sum over epochs. Predicate checks, index maintenance, and planning after rejection are additional costs and are reported separately in the experiment.

We compare full replanning, complete validation of the retained plan, dependency-indexed validation, and unchecked reuse under the same planner, action model, RGB observations, and generated update schedule. The study generated 800 layouts and retained 513 by a method-independent eligibility rule, yielding 12,312 method–condition rows and 3,078 complete paired cells. Every stored environment trace replays exactly.

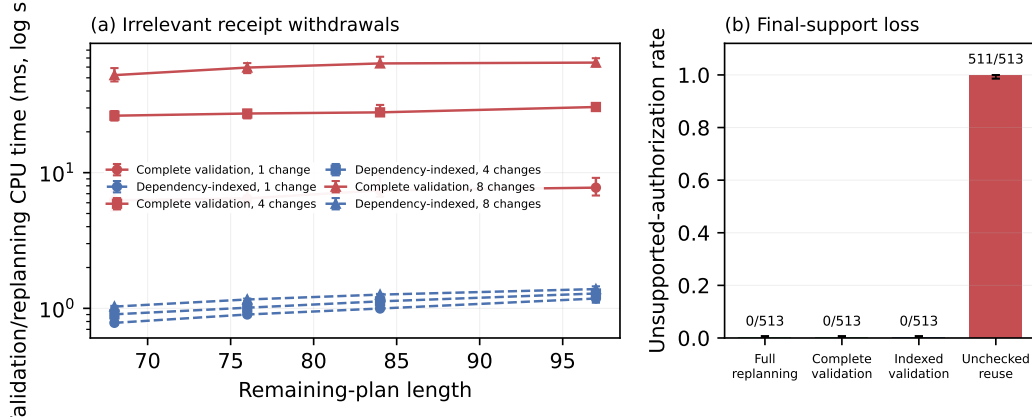


Figure 2: Two distinct interventions under matched RGB access. (a) In the primary long-plan population with four or eight irrelevant receipt withdrawals, dependency-indexed and complete retained-plan validation agree on all 1326 paired decisions, while indexing reduces validation-and-replanning CPU time by 96.7% (layout-bootstrap 95% interval [96.4%, 96.9%]); curves also show the one-withdrawal condition descriptively. (b) After final-support loss, every checking method prevents unsupported authorization, whereas unchecked reuse does not. Binomial bars are exact 95% intervals.

282 The prospectively frozen primary population contains 221 layouts with at least 64 remaining ac-
 283 tions and four or eight distinct irrelevant receipt withdrawals, yielding 1326 paired timing cells.
 284 Dependency-indexed and complete retained-plan validation agree on 100.0% of authorization de-
 285 cisions. We define *validation-and-replanning CPU time* as index construction, index maintenance,
 286 required-claim validation, and planner calls; it excludes CNN inference, simulator stepping, and
 287 action execution. Dependency indexing reduces this endpoint by 96.7% relative to complete vali-
 288 dation (layout-bootstrap 95% interval [96.4%, 96.9%]). Panel (a) additionally shows the prospectively
 289 specified one-update condition descriptively.

290 In a separate final-support-loss intervention, full replanning, complete validation, and dependency-
 291 indexed validation make zero unsupported authorizations in 513 episodes; unchecked reuse makes
 292 511. Thus the efficiency result concerns sparse irrelevant changes, while the safety result tests removal
 293 of the final support for the current action. Neither comparison asserts an advantage in physical action
 294 count or perception accuracy.

295 7 Discussion

296 The experiments establish three separable facts. First, target granularity and the resulting visible
 297 target prevalence strongly affect witness acquisition under fixed wiring, visibility, and action budgets.
 298 Second, no candidate ordering is uniformly best: random branch diversification is competitive on
 299 two graphs, while learned second-hop ordering helps on HemiBrain. Third, certificate validity is
 300 independent of search quality. A weak policy may return no certificate; a successful policy is accepted
 301 only when its active receipts support every required claim.

302 The RGB instantiation isolates a control mechanism rather than a broad robotics result. On long
 303 retained plans with repeated sparse updates, dependency indexing preserves the decisions of complete
 304 retained-plan validation while reducing validation-and-replanning CPU time, including index con-
 305 struction and maintenance. Unchecked reuse demonstrates the corresponding safety failure after final
 306 support loss. The RGB study evaluates evidence management rather than comparative performance
 307 against model-free reinforcement learning, physical action efficiency, or the correctness of learned
 308 perception. Only the interface transfers between graph exploration and RGB planning; policies,
 309 planners, and required-claim definitions remain task-specific.

310 The guarantees require a trusted receipt issuer, complete retained-plan dependencies, and declared
 311 update semantics; they neither authenticate cryptographically nor detect a systematically wrong issuer.
 312 Selective and complete retained-plan validation are equivalent under Proposition 3, but neither need
 313 match global replanning after an unrelated world change.

References

- [1] Devendra Singh Chaplot, Dhiraj Gandhi, Saurabh Gupta, Abhinav Gupta, and Ruslan Salakhutdinov. Learning to explore using active neural SLAM. In *International Conference on Learning Representations*, 2020.
- [2] Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo Perez-Vicente de Lazcano, Lucas Willems, Salem Lahlou, Suman Pal, Pablo Samuel Castro, and Jordan Terry. MiniGrid & MiniWorld: Modular & customizable reinforcement learning environments for goal-oriented tasks. *CoRR*, abs/2306.13831, 2023. URL <https://arxiv.org/abs/2306.13831>.
- [3] Jody Clements, Tom Dolafi, Lowell Umayam, Nicole L. Neubarth, Stuart Berg, Louis K. Scheffer, and Stephen M. Plaza. neuPrint: Analysis tools for EM connectomics. *bioRxiv*, 2020. doi: 10.1101/2020.01.16.909465.
- [4] Patrick R. Conrad and Brian C. Williams. Drake: An efficient executive for temporal plans with choice. *Journal of Artificial Intelligence Research*, 42:607–659, 2011. doi: 10.1613/JAIR.3478.
- [5] Rajarshi Das, Shehzaad Dhuliawala, Manzil Zaheer, Luke Vilnis, Ishan Durugkar, Akshay Krishnamurthy, Alex Smola, and Andrew McCallum. Go for a walk and arrive at the answer: Reasoning over paths in knowledge bases using reinforcement learning. In *International Conference on Learning Representations*, 2018.
- [6] Amol Deshpande, Lisa Hellerstein, and Devorah Kletenik. Approximation algorithms for stochastic boolean function evaluation and stochastic submodular set cover. In *Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms*, 2014. URL <https://www.cs.umd.edu/projects/reucaar/approx.pdf>.
- [7] Jon Doyle. A truth maintenance system. *Artificial Intelligence*, 12(3):231–272, 1979. doi: 10.1016/0004-3702(79)90008-0.
- [8] Mikhail Galkin, Jincheng Zhou, Bruno Ribeiro, Jian Tang, and Zhaocheng Zhu. A foundation model for zero-shot logical query reasoning. In *Advances in Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=JRSyMBBJi6>.
- [9] Roman Garnett, Yamuna Krishnamurthy, Xuehan Xiong, Jeff Schneider, and Richard Mann. Bayesian optimal active search and surveying. In *Proceedings of the 29th International Conference on Machine Learning*, 2012.
- [10] Caelan Reed Garrett, Rohan Chitnis, Rachel Holladay, Beomjoon Kim, Tom Silver, Leslie Pack Kaelbling, and Tomás Lozano-Pérez. Integrated task and motion planning. *Annual Review of Control, Robotics, and Autonomous Systems*, 4:265–293, 2021. doi: 10.1146/annurev-control-091420-084139.
- [11] Timothy Lebo, Satya Sahoo, and Deborah McGuinness. PROV-O: The PROV ontology. W3c recommendation, World Wide Web Consortium, 2013. URL <https://www.w3.org/TR/prov-o/>.
- [12] Xiaotian Liu, Ali Pesaraghader, Jaehong Kim, Tanmana Sadhu, Hyejeong Jeon, and Scott Sanner. ActiveVOO: Value of observation guided active knowledge acquisition for open-world embodied lifted regression planning. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/995c4e28ac35b90ca841053f80d2f1f8-Abstract-Conference.html.
- [13] George C. Necula. Proof-carrying code. In *Proceedings of the 24th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 106–119, 1997. doi: 10.1145/263699.263712.
- [14] Hongyu Ren, Weihua Hu, and Jure Leskovec. Query2Box: Reasoning over knowledge graphs in vector space using box embeddings. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=BJgr4kSFDS>.

- 361 [15] Louis K. Scheffer, C. Shan Xu, Michal Januszewski, Zhiyuan Lu, Shin-ya Takemura, Kenneth J.
 362 Hayworth, Gary B. Huang, Kazunori Shinomiya, Jeremy Maitin-Shepard, et al. A connectome
 363 and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020. doi: 10.7554/eLife.
 364 57443.
- 365 [16] Alexander Shapson-Coe, Michal Januszewski, Daniel R. Berger, Art Pope, Yuelong Wu, Tim
 366 Blakely, Richard L. Schalek, Peter H. Li, et al. A petavoxel fragment of human cerebral cortex
 367 reconstructed at nanoscale resolution. *Science*, 384(6696):eadk4858, 2024. doi: 10.1126/
 368 science.adk4858.
- 369 [17] I-Hsiang Shu, Robert T. Effinger, and Brian C. Williams. Enabling fast flexible planning
 370 through incremental temporal reasoning with conflict extraction. In *Proceedings of the Fifteenth*
 371 *International Conference on Automated Planning and Scheduling*, pages 252–261, 2005. URL
 372 <https://cdn.aaai.org/ICAPS/2005/ICAPS05-026.pdf>.
- 373 [18] Shin-ya Takemura, Kenneth J. Hayworth, Gary B. Huang, Michal Januszewski, Zhiyuan Lu,
 374 Elizabeth C. Marin, Stephan Preibisch, C. Shan Xu, et al. A connectome of the male *Drosophila*
 375 ventral nerve cord. *eLife*, 13:RP97769, 2024. doi: 10.7554/eLife.97769.
- 376 [19] Paul Zaidins, Robert P. Goldman, Ugur Kuter, Dana Nau, and Mark Roberts. HTN plan repair
 377 algorithms compared: Strengths and weaknesses of different methods. In *Proceedings of the*
 378 *Thirty-Fifth International Conference on Automated Planning and Scheduling*, volume 35, pages
 379 297–305, 2025. doi: 10.1609/icaps.v35i1.36131.